

Basic Electronics

Tutorial 5

PN junction diode and applications

Semiconductor Diode

1. a. Determine the thermal voltage for a diode at a temperature of 20°C.
 b. For the same diode of part (a), find the diode current using $I_D = I_s \left(e^{\frac{V_d}{nV_T}} - 1 \right)$ if $I_s = 40 \text{ nA}$, $n = 2$ (low value of V_D), and the applied bias voltage is 0.5 V.
2. a. Using Eq. $I_D = I_s \left(e^{\frac{V_d}{nV_T}} - 1 \right)$, determine the diode current at 20°C for a silicon diode with $n = 2$, $I_s = 0.1 \text{ mA}$ at a reverse-bias potential of -10 V.
 b. Is the result expected? Why?
3. Given a diode current of 6 mA, $V_T = 26 \text{ mV}$, $n = 1$, and $I_s = 1 \text{ nA}$, find the applied voltage V_D .
4. In the reverse-bias region the saturation current of a silicon diode is about $0.1 \text{ } \mu\text{A}$ ($T = 20^\circ\text{C}$). Determine its approximate value if the temperature is increased 40°C .
5. Determine the forward voltage drop across the diode whose characteristics appear in Fig. 1 at temperatures of -75°C , 25°C , 125°C and a current of 10 mA. For each temperature, determine the level of saturation current. Compare the extremes of each and comment on the ratio of the two.

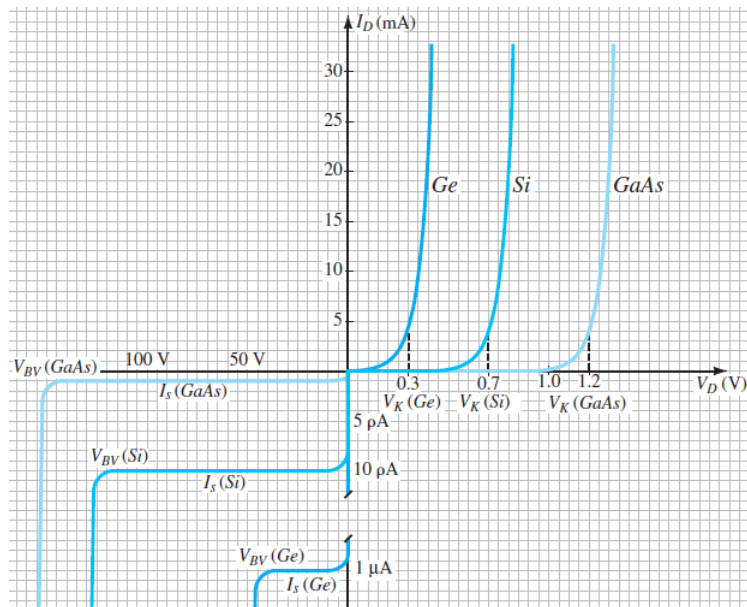


Figure 1:

Resistance Levels

- Determine the static or dc resistance of the commercially available diode of Fig. 2 at a forward currents of 4 mA and 15 mA and then compare results.

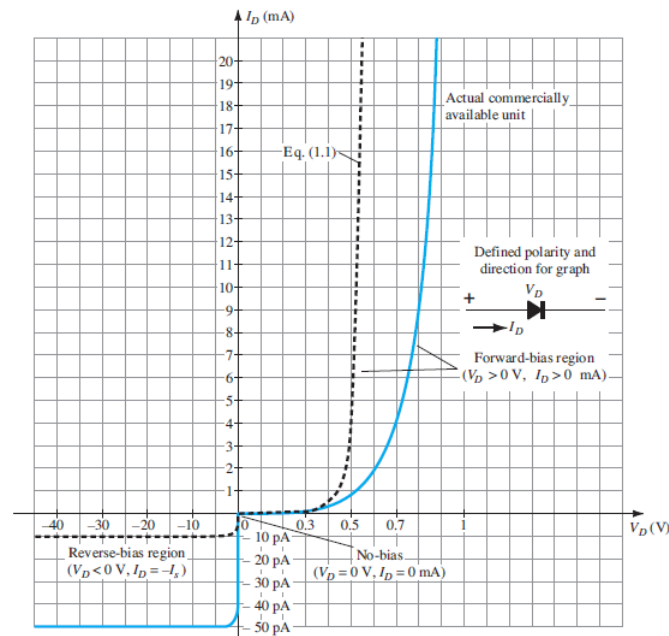


Figure 2:

- Calculate the dc and ac resistances for the diode of Fig. 2 at a forward current of 10 mA and compare their magnitudes.
- Determine the average ac resistance for the diode of Fig. 2 for the region between 0.6 V and 0.9 V. Determine the ac resistance for the same diode of at 0.75 V and compare it to the average ac resistance.

Diode Equivalent Circuits

- Find the piecewise-linear equivalent circuit for the germanium and gallium arsenide diodes of Fig. 1 .

Zener diode

- The following characteristics are specified for a particular Zener diode: $V_Z = 29$ V, $V_R = 16.8$ V, $I_{ZT} = 10$ mA, $I_R = 20$ mA, and $I_{ZM} = 40$ mA. Sketch the characteristic curve.
- Using the curves of Fig. 3 , what level of temperature coefficient would you expect for a 20-V diode? Repeat for a 5-V diode. Assume a linear scale between nominal voltage levels and a current level of 0.1 mA.

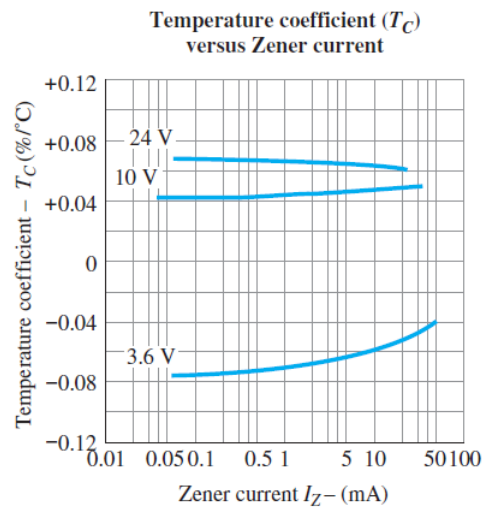


Figure 3:

12. Determine the dynamic impedance for the 24-V diode at $I_Z = 10$ mA for Fig. 4 . Note that it is a log scale.

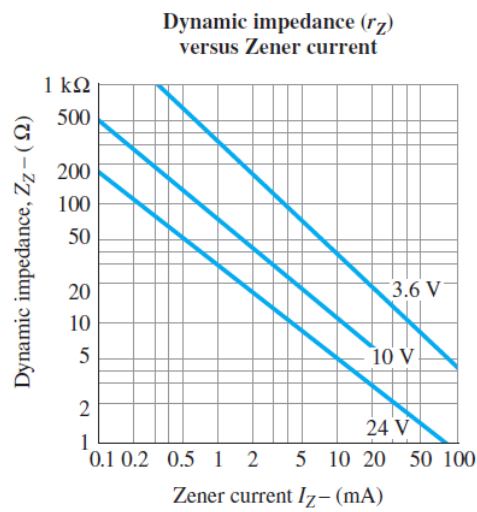


Figure 4: